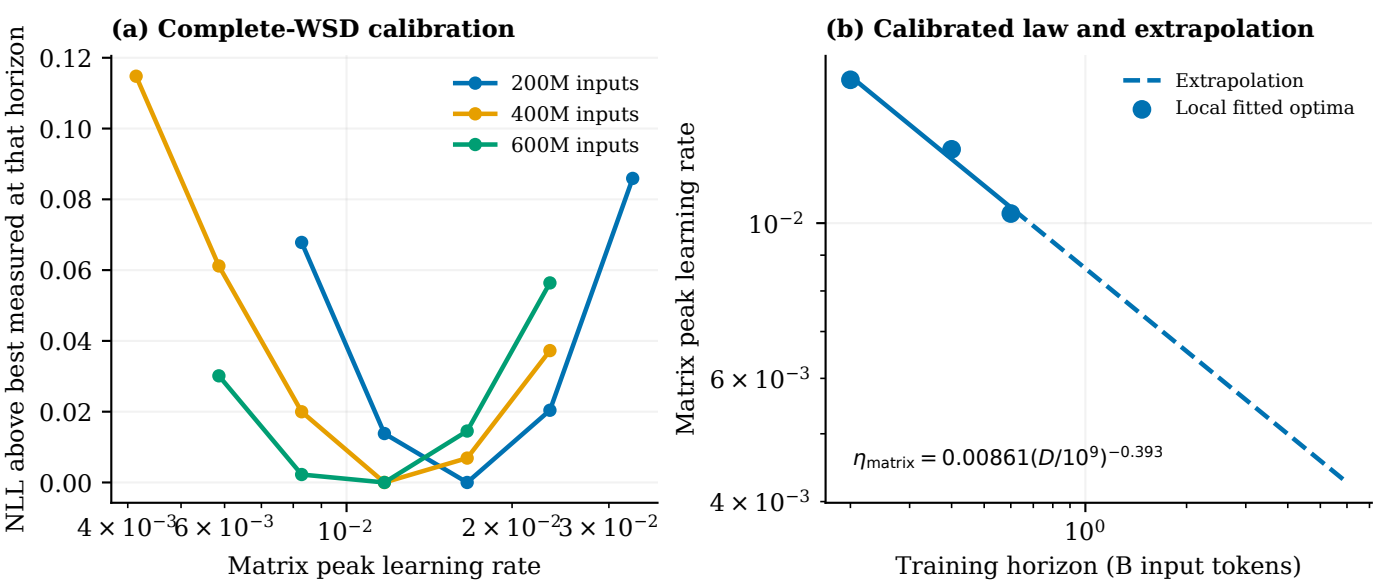
(a) Complete-WSD calibration
200M inputs
400M inputs
600M inputs
NLL above best measured at that horizon
0.12
0.10
0.08
0.06
0.04
0.02
0.00
4×10^{-3}
6×10^{-3}
10^{-2}
2×10^{-2}
3×10^{-2}
Matrix peak learning rate
(b) Calibrated law and extrapolation
Extrapolation
Local fitted optima
Matrix peak learning rate
10^{-2}
6×10^{-3}
4×10^{-3}
$\eta_{\text{matrix}} = 0.00861(D/10^9)^{-0.393}$
10^0
Training horizon (B input tokens)